

BAKALY, Russian Federation

WMO: 286150

Lat: 55.183N

Lon: 53.790E

Elev: 125

StdP: 99.83

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.8	-33.5	0.2	-30.9	-30.3	0.2	-27.8	12.5	-11.2	11.6	-9.6	1.3	350	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.7	19.8	28.8	19.3	27.0	18.3	21.2	28.4	20.3	27.0	19.4	25.2	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.8	23.8	17.9	13.0	23.1	17.1	12.4	22.1	61.7	28.3	58.9	27.3	55.6	25.3	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	9.8	8.5	-35.6	33.3	3.1	1.6	-37.9	34.5	-39.7	35.4	-41.5	36.3	-43.7	37.5		
(5)			-35.7	22.8	3.1	1.2	-38.0	23.7	-39.8	24.4	-41.5	25.0	-43.8	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-11.6	-11.6	-5.2	4.5	12.5	17.0	19.2	16.5	11.5	5.0	-4.5	-11.2		
	DBStd	12.79	7.59	8.40	5.09	5.88	4.97	4.19	3.52	4.19	4.64	4.82	6.34	8.10		
	HDD10.0	3258	668	606	472	183	32	2	0	1	36	167	435	657		
	HDD18.3	5518	927	839	730	414	189	75	33	88	211	412	685	916		
	CDD10.0	921	0	0	0	19	109	211	286	202	80	13	0	0		
	CDD18.3	138	0	0	0	0	8	34	61	31	5	0	0	0		
	CDH23.3	1519	0	0	0	8	131	367	640	314	59	0	0	0		
	CDH26.7	458	0	0	0	1	22	118	193	108	15	0	0	0		
(14) Wind	WSAvg	3.3	3.5	3.8	3.7	3.3	3.7	2.9	2.6	2.8	2.9	3.8	3.6	3.6		
(15) Precipitation	PrecAvg	464	27	20	20	21	36	57	44	53	38	48	34	31		
	PrecMax	623	54	40	39	47	73	94	128	159	86	98	77	57		
	PrecMin	268	0	0	2	1	8	16	4	11	4	6	1	3		
	PrecStd	99	13	11	13	15	20	25	32	32	24	27	23	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	2.7	6.6	24.4	29.1	33.0	32.7	32.7	29.3	20.4	6.4	2.0		
		MCWB	0.8	1.3	4.2	14.6	16.2	21.0	21.2	19.9	17.3	11.6	4.8	0.9		
	2%	DB	0.7	1.5	4.1	21.1	26.6	29.9	30.7	29.9	24.7	17.8	5.0	1.2		
		MCWB	0.1	0.5	2.0	11.9	15.9	19.8	20.4	19.4	15.1	11.4	3.7	0.6		
	5%	DB	-0.3	0.6	2.6	18.5	24.5	27.6	28.8	26.7	22.0	15.1	3.6	0.4		
		MCWB	-0.9	-0.2	1.2	10.6	15.3	18.9	20.0	18.2	14.4	9.6	2.6	-0.1		
	10%	DB	-1.5	-0.7	1.6	14.9	21.9	25.1	27.1	24.2	19.3	12.6	2.5	-0.8		
		MCWB	-2.2	-1.5	0.4	9.0	14.1	17.9	19.2	17.2	13.5	9.3	1.5	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	1.4	5.0	14.5	17.7	22.3	22.4	21.5	18.6	13.9	5.2	1.2		
		MCDB	1.4	2.1	6.9	24.0	25.4	30.0	29.0	28.3	25.8	17.6	5.9	1.6		
	2%	WB	0.2	0.6	2.3	12.4	16.7	20.8	21.4	20.1	16.6	12.4	3.9	0.6		
		MCDB	0.6	1.4	3.5	20.0	24.5	28.2	28.5	27.1	21.9	15.5	4.8	1.1		
	5%	WB	-0.9	-0.2	1.4	11.0	15.7	19.7	20.7	19.0	15.2	10.7	2.9	-0.1		
		MCDB	-0.4	0.6	2.5	17.6	23.5	25.9	27.5	25.2	20.4	14.6	3.5	0.4		
	10%	WB	-2.2	-1.5	0.6	9.0	14.7	18.5	19.9	17.8	13.9	9.1	1.7	-1.3		
		MCDB	-1.6	-0.8	1.5	14.7	21.0	24.0	25.9	22.8	18.9	12.4	2.2	-0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.8	8.4	9.7	11.1	10.7	11.2	10.3	9.5	6.4	5.6	6.9		
		MCDBR	6.4	4.7	6.9	15.5	15.3	14.4	14.2	14.6	14.0	12.8	5.7	6.1		
		MCWBR	6.4	4.4	5.6	7.6	6.7	6.0	5.3	5.5	6.6	7.3	4.5	6.0		
		MCDBR	6.3	4.9	5.7	14.6	14.0	12.9	13.0	12.7	11.9	11.5	4.1	6.0		
	5% WB	MCWBR	6.3	4.7	4.9	7.8	6.5	5.8	5.3	5.3	6.9	7.1	4.1	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.319	0.371	0.413	0.399	0.394	0.411	0.433	0.381	0.354	0.317	0.292		
	taud		2.042	2.030	1.960	2.092	2.272	2.336	2.290	2.206	2.323	2.283	2.183	2.021		
	Ebn at Noon		590	724	770	801	843	852	827	777	772	692	570	488		
	Edh at Noon		68	100	137	139	123	117	121	124	96	79	60	56		
(44) All-Sky Solar Radiation	RadAvg		0.81	1.83	3.27	4.40	5.42	5.89	5.80	4.55	2.88	1.31	0.57	0.46		
	RadStd		0.10	0.17	0.32	0.44	0.50	0.57	0.53	0.41	0.34	0.29	0.12	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page